

An integral degree-three Gersten counterexample

OpenAI

September 26, 2026

Abstract

We construct a two-dimensional ramified regular local ring A in mixed characteristic $(0, 5)$ for which the integral map $K_3(A) \rightarrow K_3(\text{Frac } A)$ has nonzero kernel. This gives a negative answer to the unrestricted integral Gersten conjecture.

1 Introduction

For a regular local domain B with fraction field F_B , the integral Gersten conjecture asks whether, for each $n \geq 0$, the augmented complex of Quillen groups

$$0 \longrightarrow K_n(B) \longrightarrow K_n(F_B) \longrightarrow \bigoplus_{\text{ht } \mathfrak{p}=1} K_{n-1}(\kappa(\mathfrak{p})) \longrightarrow \cdots$$

is exact. Here $\kappa(\mathfrak{p})$ is the residue field at a prime, and the later maps are the codimension-residue maps. In particular, exactness requires the first map $K_n(B) \rightarrow K_n(F_B)$ to be injective. We construct a ring for which this requirement fails in degree three.

Fix a primitive fifth root of unity ζ , and define

$$L = \mathbb{Q}_5(\zeta), \quad V = \mathcal{O}_L, \quad \pi = 1 - \zeta.$$

Let $\mathcal{X} \subset \mathbb{P}_V^2$ be the projective hypersurface

$$(X - Z)^5 - XY^4 + Y^5 + \zeta Z^5 = 0.$$

On its special fiber take $m = [0 : 0 : 1]$, and set $A = \mathcal{O}_{\mathcal{X}, m}$. We use G_i for coherent K -theory and $G_i(-; \mathbb{Z}/5)$ for its mod-five coefficient groups.

Theorem 1.1. *The ring A is a two-dimensional Noetherian regular local domain of mixed characteristic $(0, 5)$, with $\pi \in \mathfrak{m}_A^2$. There is an integral class $c \in K_3(A)$ such that*

$$c|_{\text{Frac } A} = 0, \quad c|_{A[1/\pi]} \bmod 5 \neq 0 \quad \text{in } G_3(A[1/\pi]; \mathbb{Z}/5).$$

In particular, $K_3(A) \rightarrow K_3(\text{Frac } A)$ is not injective.

Thus the unrestricted integral Gersten complex can fail already at its initial injection. The class c is not in $5K_3(A)$: any such multiple would have zero coefficient reduction, and hence zero reduction after restriction to $A[1/\pi]$. This conclusion does not determine the order of c or its image in $K_3(A) \otimes \mathbb{Q}$.

Context and earlier results

Gersten asked, among other questions, whether the transfer from the residue field to a discrete valuation ring vanishes [5, Problem 9]. Localization connects this question to injectivity of restriction to the fraction field: the image of the residue-field transfer is its kernel. Quillen formulated the general regular-local conjecture and proved it for regular semilocalizations of finite-type algebras over a field [11, §7, Conjecture 5.10 and Theorem 5.11]. Sherman established the equicharacteristic DVR case [12, p. 498], and Panin proved the full equicharacteristic case [9, Theorem A]. These results place the remaining integral question in mixed characteristic.

In mixed characteristic, smoothness over a DVR and the choice of coefficients lead to different positive results. Gillet and Levine’s relative theory reduces the smooth-over-DVR setting to the DVR case [6]. Geisser and Levine proved finite-coefficient Gersten exactness for semilocal rings essentially smooth over a DVR, including coefficients of residue-characteristic order [4, Theorem 8.2]. Druzhinin’s preprint states exactness for essentially smooth local Henselian schemes over a DVR [1, Theorem 4.5]. A separate result of Skalit gives finite-coefficient exactness for every unramified regular local ring [13, Corollary 5.7]; in mixed characteristic $(0, p)$, unramified here means $p \in \mathfrak{m} \setminus \mathfrak{m}^2$. His integral result is a reduction to DVRs essentially smooth over $\mathbb{Z}$ [13, Theorem 5.4]. Our ring satisfies $5 \in \mathfrak{m}_A^2$, and its special fiber over the chosen V is singular at m ; it lies outside the unramified and smooth-over- V settings just described.

The distinction between integral and coefficient classes governs the proof. A finite-coefficient kernel class may have a nonzero coefficient boundary and hence fail to lift integrally. Even when it lifts, its integral generic restriction may be nonzero but divisible by the coefficient prime. Feld’s mod-three degree-two example and its integral divisibility corollary illustrate this second issue [2, Theorem 1.1 and Corollary 5.1].

Feld’s construction also provides a methodological antecedent: push a Bott-type class from a divisor and detect its image after a flat cyclotomic base change and inversion of three. The symbol sketch in [2, Remark 5.3, footnote 3] gives a related use of symbols. Here we arrange the integral lift on the divisor itself, so support proves integral generic vanishing before the norm test is applied. No result from Feld’s or Druzhinin’s preprint is an input to this proof.

The companion paper *An integral counterexample to Gersten’s conjecture* [8, Theorem 1.1] gives an integral degree-five counterexample at $A_5 = (V_0[x, y]/(5 + x^4 + y^4))_{(5, x, y)}$, where V_0 is the ring of integers of the unramified degree-eight extension of $\mathbb{Q}_5$. Its integral cyclotomic lift and Picard-valued specialization test form a different proof route. Here the quintic model supports a Steinberg lift and a norm valuation test, proved entirely within this paper. The two results concern different rings and do not determine the smallest degree in which integral Gersten injectivity can fail.

The route through the proof

Write $C = \mathcal{X}_L$ for the projective generic curve. On the chart $Z \neq 0$ let $x = X/Z$ and $y = Y/Z$. The divisor $y = 0$ in $\text{Spec } A$ has ring $S = A/yA$. Section 2 proves that S is a DVR with uniformizer $t = x \bmod y$, and that its fraction field D is a totally ramified degree-five extension of L satisfying

$$N_{D/L}(t) = \pi, \quad (1 - t)^5 = \zeta.$$

For a finite extension E/L , write k_E for the residue field of its valuation ring; its residue degree is $[k_E : \mathbb{F}_5]$. The corresponding closed point $q_0 \in C$ has field $\kappa(q_0) = D$ and specializes to m , with

$[D : L] = 5$ but $[k_D : \mathbb{F}_5] = 1$. Every closed point q specializing elsewhere satisfies $5 \mid [k_{\kappa(q)} : \mathbb{F}_5]$. This difference in residue degrees supplies the nonvanishing test.

Write $[u]$ for the integral K_1 class of a nonzero field element. Section 3 chooses a class $\beta \in K_2(L; \mathbb{Z}/5)$ whose coefficient boundary in $K_1(L)$ is $[\zeta]$ and constructs an integral lift of $a_D = \beta_D \cdot [t]$ in two steps, where β_D is the restriction of β to D . The identity $(1 - t)^5 = \zeta$ turns the coefficient boundary of a_D into a multiple of the Steinberg symbol $\{1 - t, t\} = 0$ in $K_2(D)$ [7]. Hence a_D lifts to an integral class in $K_3(D)$. Quillen's finite-field calculation gives $K_2(\mathbb{F}_5) = 0$ [10], so localization then lifts this integral class to $K_3(S)$. Pushforward from S produces $c \in K_3(A)$, and restriction to $\text{Frac } A$ kills it because its support is the divisor $y = 0$.

To detect this candidate, Section 4 proves that for every finite extension E/L , multiplication by the restriction β_E identifies $E^\times/E^{\times 5}$ with $K_3(E; \mathbb{Z}/5)$. This calculation combines the multiplicative motivic-to- K spectral sequence of Friedlander and Suslin, Suslin's product-preserving comparison, and the norm-residue comparison [3, 14, 15]. The projection formula identifies field transfer with norm. Normalized valuation modulo five consequently kills every transferred coefficient class from a field of residue degree divisible by five, whereas a_D has transferred valuation one because $N_{D/L}(t) = \pi$.

Section 5 proves that this integral class is nonzero. The scheme $T = \text{Spec } A[1/\pi]$ is an inverse limit of affine opens of C with flat transition maps. These opens contain every closed point specializing to m , so their omitted points have residue degree divisible by five. If the coefficient reduction of $c|_T$ vanished, continuity of coherent K -theory would make it vanish on one such open. Localization would then express the supported coefficient class $(q_0)_*(\beta_D \cdot [t])$ as a sum of classes supported at omitted points. Proper pushforward to L and the norm valuation give zero for every term on that side and one for q_0 , a contradiction. We work in coherent G -theory on C , which need not be smooth.

2 The model and its distinguished divisor

We verify the local ring in Theorem 1.1 and construct the divisor on which the integral class will be lifted. The special fiber forces a factor of five in residue degrees away from m , while the divisor through m has residue degree one and a uniformizer of norm π . These properties will distinguish its class from classes whose supporting points specialize away from m . For a finite extension E/L , write $\mathcal{O}_E$ for its valuation ring and k_E for its residue field.

Lemma 2.1. *The generic fiber $C = \mathcal{X}_L$ is an integral projective curve. The ring $A = \mathcal{O}_{\mathcal{X}, m}$ is a two-dimensional Noetherian regular local domain of mixed characteristic $(0, 5)$ with $\pi \in \mathfrak{m}_A^2$. For every closed point $q \in C$ with residue field $E = \kappa(q)$, the E -point extends to $\text{Spec } \mathcal{O}_E \rightarrow \mathcal{X}$. If its closed image is not m , then $5 \mid [k_E : \mathbb{F}_5]$.*

On the chart $Z \neq 0$, set $x = X/Z$, $y = Y/Z$, and let t be the image of x in $S = A/yA$. Then

$$S \cong V[t]/((t - 1)^5 + \zeta) \tag{1}$$

is a DVR with uniformizer t . Its fraction field D is totally ramified of degree five over L , and

$$N_{D/L}(t) = \pi, \quad \eta = 1 - t, \quad \eta^5 = \zeta. \tag{2}$$

The equation $Y = 0$ cuts out a single closed point $q_0 \in C$ with residue field D .

Proof. The polynomial

$$\Phi_5(1 - T) = T^4 - 5T^3 + 10T^2 - 10T + 5$$

is Eisenstein at five. Therefore $\pi = 1 - \zeta$ is a uniformizer of V , its residue field is $\mathbb{F}_5$, and $5 = u\pi^4$ for some $u \in V^\times$. The homogeneous equation P defining $\mathcal{X}$ reduces to

$$\overline{P} = X^5 - XY^4 + Y^5.$$

If α is a root of $s^5 - s + 1$, then Frobenius sends α to $\alpha - 1$. Its orbit has length exactly five: the r th iterate is $\alpha - r$, which first returns at $r = 5$. Thus $s^5 - s + 1$ is irreducible over $\mathbb{F}_5$, and so is its homogenization $\overline{P}$. This remains irreducible on adjoining the independent variable Z . A factorization of the primitive homogeneous polynomial P over L would, by Gauss' lemma, give primitive positive-degree homogeneous factors over V and hence a factorization of $\overline{P}$. Consequently C is integral. It is a projective plane curve, so it has dimension one.

Put

$$F = (x - 1)^5 - xy^4 + y^5 + \zeta, \quad R = V[x, y]/(F), \quad \mathfrak{m} = (\pi, x, y) \subset R.$$

In the regular local ring $V[x, y]_{(\pi, x, y)}$ of dimension three, $F \equiv -\pi$ modulo the square of the maximal ideal. Thus quotienting by F gives the regular local ring $A = R_{\mathfrak{m}}$ of dimension two. A regular local ring is a domain. Since F is monic in x , the ring R is free over $V[y]$ and hence flat over V . Its localization A is also V -flat, so V embeds into A . Together with $A/\mathfrak{m}_A = \mathbb{F}_5$, this proves mixed characteristic $(0, 5)$. Expanding $F = 0$ gives

$$\pi = x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - xy^4 + y^5.$$

In particular, $\pi \in (x, y)A$ and $\mathfrak{m}_A = (x, y)A$. Since 5 is divisible by π , the term $5x$ is in $(x, y)^2A$, as are all remaining terms on the right. Hence $\pi \in \mathfrak{m}_A^2$. Since $5 = u\pi^4$, also $5 \in \mathfrak{m}_A^2$, so A is ramified in the usual mixed-characteristic sense. The reduction of the equation has no linear term at m , so the one-dimensional special fiber is singular there; A is not smooth over V .

Modulo y , the equation becomes

$$f(t) = (t - 1)^5 + \zeta = t^5 - 5t^4 + 10t^3 - 10t^2 + 5t - \pi.$$

This is Eisenstein at π . The finite free V -algebra $B = V[t]/(f)$ is a domain. Every maximal ideal lies over the maximal ideal of V , and $B/\pi B = \mathbb{F}_5[t]/(t^5)$, so B is local. Since $f(t) = 0$ implies $\pi \in (t)B$, its maximal ideal is (t) . Its dimension is one, making it a DVR. The localization defining A/yA leaves this local ring unchanged, proving (1). Its fraction field has degree five over L and residue degree one, hence total ramification. The constant term gives

$$N_{D/L}(t) = (-1)^5 f(0) = \pi.$$

The relation $(1 - t)^5 = \zeta$ also follows from $f(t) = 0$. As $S[1/\pi] = D$, the equation $y = 0$ on the generic affine chart gives a single closed point. There is no additional point with $Y = Z = 0$ on C , since substitution in P would force $X = 0$. This proves the assertion about q_0 on all of C .

We have constructed the distinguished point with residue degree one. It remains to establish the residue-degree constraint on points that specialize away from m . A closed point of C has residue field finite over L . Properness of $\mathcal{X}/V$ extends its E -point to $\text{Spec } \mathcal{O}_E$. On the special fiber $Y = 0$ forces $X = 0$, so m is the only point with $Y = 0$. If the closed image is elsewhere, X/Y in k_E is a root of $s^5 - s + 1$. Thus k_E contains $\mathbb{F}_{5^5}$, and $5 \mid [k_E : \mathbb{F}_5]$. $\square$

3 Lifting on the divisor

The divisor $\text{Spec } S \subset \text{Spec } A$ now supplies an integral class with zero generic restriction. We first lift a coefficient class to $K_3(D)$ and then extend that lift across the closed point of S . Its coefficient image will be tested for nonvanishing only after this construction.

Let M_5 be the mod-five Moore spectrum. We use $K_i(B; \mathbb{Z}/5) = \pi_i(K(B) \wedge M_5)$, and the same convention for G -theory. The coefficient sequence is

$$K_i(B) \xrightarrow{5} K_i(B) \xrightarrow{\rho_B} K_i(B; \mathbb{Z}/5) \xrightarrow{\partial_{\text{coef}}} K_{i-1}(B) \xrightarrow{5} K_{i-1}(B). \quad (3)$$

Here ρ_B denotes coefficient reduction, and ∂_{coef} denotes the connecting map. Multiplication of a coefficient class by an integral class uses the natural $K(B)$ -module structure on $K(B) \wedge M_5$.

For an element u of a field's multiplicative group, write $[u]$ for its integral K_1 class. Since $[\zeta] \in K_1(L) = L^\times$ has order five, exactness permits us to fix

$$\beta \in K_2(L; \mathbb{Z}/5), \quad \partial_{\text{coef}} \beta = [\zeta].$$

For every finite extension E/L , write β_E for its restriction. The class to be lifted is

$$a_D = \beta_D \cdot [t] \in K_3(D; \mathbb{Z}/5).$$

Resolution identifies K -theory with coherent G -theory for the regular rings S and A ; closed-immersion pushforward is taken in G -theory [11, §7.1 and §7, paragraph 2.7 and the discussion following (2.8)].

Proposition 3.1. *There exists $b \in K_3(S)$ whose restriction to D , followed by coefficient reduction, is a_D . If $j : \text{Spec } S \hookrightarrow \text{Spec } A$ and $c = j_* b \in G_3(A) = K_3(A)$, then $c|_{\text{Frac } A} = 0$. Moreover, for $T = \text{Spec } A[1/\pi]$ and $i : \text{Spec } D \hookrightarrow T$,*

$$c|_T \bmod 5 = i_* a_D \quad \text{in } G_3(T; \mathbb{Z}/5).$$

Proof. The coefficient cofiber sequence is a sequence of $K(D)$ -modules. Its connecting map is therefore compatible with multiplication by an integral $K(D)$ class, up to the suspension sign. By (2), $\zeta = \eta^5$ and $t = 1 - \eta$. Both η and $1 - \eta$ are nonzero: the first has fifth power $\zeta \neq 0$, and the second is the uniformizer t of S . Matsumoto's Steinberg relation [7, §5, Lemma 5.6, Theorem 5.10, and Corollary 5.11], with the identification of symbols and products in Quillen $K_2(D)$ [16, III.6.1, IV.1.7.1, IV.1.10.1, and IV.7.2], gives

$$\partial_{\text{coef}} a_D = \pm[\zeta] \cdot [t] = \pm 5\{\eta, 1 - \eta\} = 0. \quad (4)$$

The sign does not affect the vanishing. Exactness of (3) yields an integral lift $\tilde{a}_D \in K_3(D)$ of a_D . The boundary $[\zeta] \in K_1(D)[5]$ of β_D itself is still nonzero, even though $\zeta = \eta^5$. Being a fifth power makes the class of ζ vanish in $D^\times/D^{\times 5}$, not in the integral group $K_1(D) = D^\times$. The Steinberg relation is what kills the boundary of the product a_D .

Localization and dévissage for the DVR S give [11, §7, Proposition 3.2]

$$K_3(S) \longrightarrow K_3(D) \longrightarrow K_2(\mathbb{F}_5).$$

The last group is zero by Quillen's finite-field computation [10]. We may therefore choose $b \in K_3(S)$ restricting to $\tilde{a}_D$. This is a second lift, distinct from the coefficient lift to $K_3(D)$.

Let $c = j_*b \in G_3(A) = K_3(A)$. Its restriction to $\mathrm{Spec} A[1/y]$ is zero because the closed immersion j has empty pullback there. The element y is nonzero in the domain A : its quotient S has dimension one, whereas A has dimension two. Hence y becomes invertible in $\mathrm{Frac} A$, and $c|_{\mathrm{Frac} A} = 0$ integrally.

Base change of j along $T \rightarrow \mathrm{Spec} A$ is $i : \mathrm{Spec} D \hookrightarrow T$, since $S[1/\pi] = D$. Flat base change for coherent pushforward [16, V.3.7.2] here is the natural isomorphism

$$A[1/\pi] \otimes_A M \cong D \otimes_S M$$

for a finite S -module M , with both sides viewed as $A[1/\pi]$ -modules. Thus restriction of scalars commutes with localization. Naturality of coefficient reduction gives the commutative diagram

$$\begin{array}{ccc} K_3(S) & \xrightarrow{j_*} & G_3(A) \\ \downarrow & & \downarrow \\ K_3(D; \mathbb{Z}/5) & \xrightarrow{i_*} & G_3(T; \mathbb{Z}/5). \end{array}$$

Both vertical arrows mean restriction followed by coefficient reduction. The left arrow sends b to a_D , which proves the stated identity. $\square$

4 The field comparison and norm detector

Proposition 3.1 constructs an integral class c with zero generic restriction. To prove it nonzero, we will test its coefficient image by proper pushforward to L and a valuation. Localization on the projective curve will introduce arbitrary classes in $K_3(E; \mathbb{Z}/5)$, for finite extensions E/L arising as closed-point fields. We therefore need a description of all these groups that identifies field transfer with norm.

Keep the class β and its restrictions β_E fixed as in Section 3. For every finite extension E/L , let k_E be its residue field, and normalize v_E so that a uniformizer has valuation one.

Lemma 4.1 (Multiplicative field comparison). *For $L = \mathbb{Q}_5(\zeta)$ and the fixed β above, every finite extension E/L has an isomorphism*

$$\theta_E : E^\times / E^{\times 5} \xrightarrow{\sim} K_3(E; \mathbb{Z}/5), \quad u \bmod E^{\times 5} \mapsto \beta_E \cdot [u]. \quad (5)$$

The degree-three motivic edge identifies the target with $H_{\mathrm{et}}^1(E, \mu_5^{\otimes 2})$. Under this edge, the displayed map is Kummer reduction followed by cup product with the nonzero weight-one edge image $\alpha_E \in H^0(E, \mu_5)$ of β_E .

Proof. We first identify the degree-three group, then locate the image of β_E in degree two, and finally compute multiplication by β_E on integral K_1 . The strongly convergent motivic-to- K spectral sequence for a characteristic-zero field is

$$E_2^{a,b} = H^{a-b}(E, \mathbb{Z}/5(-b)) \implies K_{-a-b}(E; \mathbb{Z}/5). \quad (6)$$

It has a pairing with the integral spectral sequence that induces the integral/coefficient K -module product on the abutment [3, Theorem 16.2 and Appendix C.2–C.3]. The comparison with motivic complexes preserves products [14, Lemma 3.4 and Theorem 6.1]. Write $s = a - b$ and $j = -b$, so a term $H^s(E, \mathbb{Z}/5(j))$ has total K -degree $2j - s$. Negative weights vanish. For $j \geq 0$, the field

motivic groups vanish for $s > j$, and the norm-residue and motivic-étale comparison theorems identify them, for every $s \leq j$, with

$$H_{\text{et}}^s(E, \mu_5^{\otimes j});$$

see [15, Theorems 6.16–6.17] and [16, VI.4.1–4.2]. In particular, the groups vanish for $s < 0$. These comparisons apply because 5 is invertible in E .

The 5-cohomological dimension of a finite extension of $\mathbb{Q}_5$ is two [16, VI.§7, before Proposition 7.3]. Thus the only potentially nonzero terms satisfy $0 \leq s \leq \min(j, 2)$. For $r \geq 2$ a differential changes (s, j) to $(s + 2r - 1, j + r - 1)$, so none can join two nonzero terms. Total degree three has the single term $(s, j) = (1, 2)$, yielding the natural identification

$$K_3(E; \mathbb{Z}/5) \simeq H_{\text{et}}^1(E, \mu_5^{\otimes 2}). \quad (7)$$

In particular, this K -group is killed by five. In total degree two the weight-one quotient of the filtration is $H^0(E, \mu_5)$; the only other possible piece is $H^2(E, \mu_5^{\otimes 2})$ in weight two.

Let α_E be the weight-one image of β_E . To see that it is nonzero, restrict to an algebraic closure $\overline{E}$. The same comparison there has only $s = 0$ terms, so its total degree-two group has just the weight-one piece $H^0(\overline{E}, \mu_5)$. Matsumoto's presentation of field K_2 ([7, §5]; see also [16, III.6.1]) gives $K_2(\overline{E})/5 = 0$: every symbol $\{u, v\}$ equals $5\{r, v\}$ after choosing $r^5 = u$. The coefficient sequence therefore identifies $K_2(\overline{E}; \mathbb{Z}/5)$ with $K_1(\overline{E})[5]$ through ∂_{coef} . The restriction of β_E is nonzero, since its boundary is the primitive root $[\zeta]$. Naturality of the filtration shows that α_E has nonzero image in $H^0(\overline{E}, \mu_5)$. It is consequently a generator of $H^0(E, \mu_5)$; only this nonvanishing is needed.

It remains to determine the actual product from these filtration pieces. Integral $K_1(E) = E^\times$ has only the weight-one piece $H^1(E, \mathbb{Z}(1))$: on the diagonal $2j - s = 1$, weights $j \geq 2$ give $s > j$ and vanish, weight zero gives $H^{-1}(E, \mathbb{Z}(0)) = 0$, and negative weights vanish. The identification $\mathbb{Z}(1) \simeq \mathbb{G}_m[-1]$ identifies reduction of $[u]$ with its Kummer class in $H^1(E, \mu_5) = E^\times / E^{\times 5}$.

Write F^j for the decreasing weight filtration. The preceding calculations give

$$F^1 K_1(E) = K_1(E), \quad F^2 K_1(E) = 0, \quad F^2 K_3(E; \mathbb{Z}/5) = K_3(E; \mathbb{Z}/5), \quad F^3 K_3(E; \mathbb{Z}/5) = 0.$$

The class β_E lies in $F^1 K_2(E; \mathbb{Z}/5)$ and has image α_E in its quotient by F^2 . The mixed filtered pairing satisfies

$$F^2 K_2(E; \mathbb{Z}/5) \cdot F^1 K_1(E) \subseteq F^3 K_3(E; \mathbb{Z}/5) = 0.$$

Thus the possible weight-two part of β_E contributes nothing: the product $\beta_E \cdot [u]$ is determined by α_E and the weight-one class of $[u]$. Product compatibility identifies it, under (7), with $\alpha_E \cup [u]$. Since α_E generates the constant one-dimensional $\mathbb{F}_5$ -module μ_5 , this cup product is an isomorphism. The product factors through fifth powers because its target is killed by five and $[u^5] = 5[u]$. This proves (5) with its asserted multiplicative identification. $\square$

In particular, the coefficient class already constructed on the divisor is $a_D = \theta_D(t)$. We now relate the comparison maps for different extensions of L .

Lemma 4.2 (Norm detector). *For every finite extension E/L and every $u \in E^\times$,*

$$\text{Tr}_{E/L} \theta_E(u) = \theta_L(N_{E/L} u). \quad (8)$$

Consequently

$$\lambda_L = (v_L \bmod 5) \circ \theta_L^{-1} : K_3(L; \mathbb{Z}/5) \longrightarrow \mathbb{Z}/5$$

satisfies

$$\lambda_L(\text{Tr}_{E/L}(a)) = 0 \quad \text{for every } a \in K_3(E; \mathbb{Z}/5) \quad \text{if } 5 \mid [k_E : k_L].$$

Proof. Restriction of scalars obeys the tensor projection formula at the level of K -theory spectra [16, V.3.3.2]. Smashing its base-field factor with M_5 gives the mixed pairing identity

$$\mathrm{Tr}_{E/L}(\beta_E \cdot [u]) = \beta \cdot \mathrm{Tr}_{E/L}[u] = \beta \cdot [N_{E/L}u].$$

The second equality uses the norm description of field K_1 transfer [16, III.1.7.1 and the following paragraph]. This proves (8).

For normalized valuations, the norm of the prime ideal of E is the $[k_E : k_L]$ th power of the prime ideal of L . Hence

$$v_L(N_{E/L}u) = [k_E : k_L]v_E(u).$$

Every $a \in K_3(E; \mathbb{Z}/5)$ has the form $\theta_E(u)$ by Lemma 4.1. Applying the formula to that u proves the assertion for all a . The factor is the residue degree, not the total degree $[E : L]$. $\square$

5 Detection on the projective curve

Proposition 3.1 has constructed an integral class c with $c|_{\mathrm{Frac} A} = 0$ and coefficient image i_*a_D on

$$T = \mathrm{Spec} A[1/\pi].$$

By Lemma 4.1, $a_D = \beta_D \cdot [t] = \theta_D(t)$. To prove its pushforward i_*a_D nonzero, we extend the supported class to the projective curve C , where proper pushforward permits the norm test. The scheme T is an inverse limit of affine opens of C with flat transition maps, and need not itself be an open subset. The following lemma makes that passage explicit.

Lemma 5.1. *Let $\Sigma = R \setminus \mathfrak{m}$, where $R = V[x, y]/(F)$ and $\mathfrak{m} = (\pi, x, y)$. For $h \in \Sigma$ put*

$$U_h = \mathrm{Spec} R[1/\pi, 1/h] \subset C.$$

Then the refinement maps $U_{hk} \rightarrow U_h$ are affine and flat, and

$$T = \varprojlim_{h \in \Sigma} U_h, \quad \varprojlim_{h \in \Sigma} G_3(U_h; \mathbb{Z}/5) \xrightarrow{\sim} G_3(T; \mathbb{Z}/5). \quad (9)$$

Every U_h contains q_0 . Its complement in C is a finite set of closed points q , and their residue fields $E_q = \kappa(q)$ satisfy $5 \mid [k_{E_q} : \mathbb{F}_5]$.

Proof. The multiplicative set Σ is directed by taking products: hk gives a common refinement of h and k . Transitivity of localization gives

$$\varprojlim_{h \in \Sigma} R[1/\pi, 1/h] = R_{\mathfrak{m}}[1/\pi] = A[1/\pi].$$

Each refinement is a principal localization, so the corresponding scheme map is affine and flat. The stages and their limit are affine Noetherian schemes, hence separated. Quillen's continuity theorem, in its coherent K' conclusion [11, §7, Proposition 2.2], therefore applies to their integral G -groups.

To pass to coefficients, use the natural short exact sequence

$$0 \longrightarrow G_3(Y)/5G_3(Y) \longrightarrow G_3(Y; \mathbb{Z}/5) \longrightarrow G_2(Y)[5] \longrightarrow 0$$

for each stage $Y = U_h$ and for $Y = T$. Filtered colimits of abelian groups are exact, and thus commute with the quotient by five and the kernel of multiplication by five. Integral continuity in degrees three and two identifies the outer terms in the resulting comparison of short exact sequences. It consequently identifies the middle terms, proving the coefficient assertion in (9).

Since $h \notin \mathfrak{m}$, its image in the local divisor ring S is a unit. Thus $q_0 = \text{Spec } D$ belongs to U_h . More generally, consider a closed point $q \in C$ with residue field E whose extension $\text{Spec } \mathcal{O}_E \rightarrow \mathcal{X}$ specializes to m . The inverse image of the chart $Z \neq 0$ contains the closed point of $\text{Spec } \mathcal{O}_E$, and hence is all of $\text{Spec } \mathcal{O}_E$. On that chart the reduction of h at m is nonzero. Its pullback to $\mathcal{O}_E$ is therefore a unit, so q lies in U_h . It follows that each point omitted from U_h specializes away from m . Lemma 2.1 then gives $5 \mid [k_E : \mathbb{F}_5]$. Finally, C is an integral projective curve and U_h is nonempty, so its closed complement consists of finitely many closed points. $\square$

Proof of Theorem 1.1. Take the integral class $c \in K_3(A)$ constructed in Proposition 3.1, and put

$$w = (q_0)_* \theta_D(t) \in G_3(C; \mathbb{Z}/5).$$

Coherent G -theory provides proper pushforward even when C is singular. The point immersion is finite, and the structural map $p : C \rightarrow \text{Spec } L$ is proper with C carrying the ample line bundle $\mathcal{O}_C(1)$. Coherent pushforward and its composition law therefore apply to both maps [11, §7, paragraph 2.7 and the discussion following (2.8)].

Flat base change for the point immersion on each U_h [16, V.3.7.2], followed by (9), identifies $w|_T$ with the pushforward of $\theta_D(t)$ along $\text{Spec } D \hookrightarrow T$. By Proposition 3.1, this is precisely the coefficient reduction of $c|_T$.

Suppose that $w|_T = 0$. The element $w|_{U_1}$ represents this class in the filtered colimit (9). A zero element in that colimit becomes zero at a finite stage, so $w|_{U_h} = 0$ for some $h \in \Sigma$. Give $Z_h = C \setminus U_h$ its reduced closed structure. Apply Moore coefficients to coherent localization and dévissage at these closed points [11, §7, Proposition 3.2 and paragraph 3.4]. The resulting exact segment is

$$\bigoplus_{q \in Z_h} K_3(E_q; \mathbb{Z}/5) \longrightarrow G_3(C; \mathbb{Z}/5) \longrightarrow G_3(U_h; \mathbb{Z}/5).$$

Consequently there are classes $a_q \in K_3(E_q; \mathbb{Z}/5)$ such that

$$w = \sum_{q \in Z_h} q_* a_q. \tag{10}$$

These are arbitrary coefficient classes supplied by localization. Lemma 4.1 applies to every finite extension E_q/L , so its surjectivity writes each of them as $a_q = \theta_{E_q}(u_q)$ for some $u_q \in E_q^\times$.

Apply the proper pushforward $p_* : G_3(C; \mathbb{Z}/5) \rightarrow K_3(L; \mathbb{Z}/5)$ to (10). For every closed point, the composite $p_* q_*$ is the field transfer. Lemma 5.1 gives $5 \mid [k_{E_q} : \mathbb{F}_5]$, and Lemma 4.2 therefore gives

$$\lambda_L(p_* q_* a_q) = v_L(N_{E_q/L}(u_q)) \bmod 5 = [k_{E_q} : \mathbb{F}_5] v_{E_q}(u_q) \bmod 5 = 0.$$

Thus (10) forces $\lambda_L(p_* w) = 0$. On the other hand, the distinguished point has residue field D , and its parameter satisfies $N_{D/L}(t) = \pi$. The same transfer formula yields

$$\lambda_L(p_* w) = \lambda_L(\text{Tr}_{D/L} \theta_D(t)) = v_L(N_{D/L}(t)) \bmod 5 = v_L(\pi) \bmod 5 = 1.$$

This contradiction proves $w|_T \neq 0$. Hence the coefficient reduction of $c|_T$ is nonzero, so the integral class $c \in K_3(A)$ is itself nonzero. Its image at the fraction field is zero by Proposition 3.1; together with the properties of A in Lemma 2.1, this proves the theorem. $\square$

References

- [1] Andrei E. Druzhinin. Gersten conjecture for K -theory on Henselian schemes and φ -motivic localisation, 2025. [arXiv:2512.01923v1](#).
- [2] Niels Feld. Finite-coefficient Gersten injectivity fails in ramified mixed characteristic, 2026. [arXiv:2608.05005v2](#), September 24, 2026.
- [3] Eric M. Friedlander and Andrei Suslin. The spectral sequence relating algebraic K -theory to motivic cohomology. *Ann. Sci. Éc. Norm. Supér. (4)*, 35:773–875, 2002.
- [4] Thomas Geisser and Marc Levine. The K -theory of fields in characteristic p . *Invent. Math.*, 139:459–493, 2000.
- [5] S. M. Gersten. Problems about higher K -functors. In *Algebraic K-theory, I: Higher K-theories*, volume 341 of *Lecture Notes in Mathematics*, pages 43–56. Springer, 1973.
- [6] Henri Gillet and Marc Levine. The relative form of Gersten’s conjecture over a discrete valuation ring: The smooth case. *J. Pure Appl. Algebra*, 46:59–71, 1987.
- [7] Hideya Matsumoto. Sur les sous-groupes arithmétiques des groupes semi-simples déployés. *Ann. Sci. Éc. Norm. Supér. (4)*, 2:1–62, 1969.
- [8] OpenAI. An integral counterexample to Gersten’s conjecture. OpenAI Math Release preprint [OAI:An-Integral-Counterexample-to-Gerstens-Conjecture-September-25-2026](#), 2026.
- [9] I. A. Panin. The equicharacteristic case of the Gersten conjecture. *Trudy Mat. Inst. Steklova*, 241:169–178, 2003.
- [10] Daniel Quillen. On the cohomology and K -theory of the general linear groups over a finite field. *Ann. of Math. (2)*, 96:552–586, 1972.
- [11] Daniel Quillen. Higher algebraic K -theory: I. In *Algebraic K-theory, I: Higher K-theories*, volume 341 of *Lecture Notes in Mathematics*, pages 85–147. Springer, 1973.
- [12] C. C. Sherman. The K -theory of an equicharacteristic discrete valuation ring injects into the K -theory of its field of quotients. *Pacific J. Math.*, 74(2):497–499, 1978.
- [13] C. Skalit. Regular morphisms and Gersten’s conjecture, 2017. [arXiv:1710.00303v1](#), October 1, 2017.
- [14] A. A. Suslin. On the Grayson spectral sequence. *Trudy Mat. Inst. Steklova*, 241:218–253, 2003.
- [15] Vladimir Voevodsky. On motivic cohomology with $\mathbf{Z}/l$ -coefficients. *Ann. of Math. (2)*, 174:401–438, 2011.
- [16] Charles A. Weibel. *The K-book: An introduction to algebraic K-theory*, volume 145 of *Graduate Studies in Mathematics*. American Mathematical Society, 2013.